Blade-concept tokens

implicit-geometry-kernel
Companion to `docs/blade/concepts/taxonomy.md`

Abstract

A catalogue token names a **blade concept**, not only a station layout. W, X, and Y are the station slice; Z is span web freeze; R is transverse faces. Example: `2D-F-LC-0R`. Authoring: `blade.concepts.BladeConcept`. Spec: `implicit_geometry_kernel.catalogue`. The Markdown source remains canonical; this PDF is a printable companion with station, planform, and schematic demonstrations.

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1 Token grammar

```
[W][X]-[Y]-[Z]-[R]     e.g.  2D-F-LC-0R    2D-F-LE-10R    2D-F-PA-10R
    | |  |  |   +-- rib count (required): 0R none | 10R ten faces
    | |  |  +------ web anchoring (required): LC | LE | PA
    | |  +--------- web orientation: F flapwise | CN chord-normal
    | +------------ spar-cap family: A B C D
    +-------------- web count: 0..5
```

Zero-web cores omit [Y]: `0A-LC-0R`, `0A-LE-10R`. Bare `2D-F` and omitted `0R` are invalid.

Display form: `SystemType[W][X]-[Y]-[Z]-[R]`. `resolve_system_type("2D-F-LC-0R")` yields a frozen `SystemLayoutSpec`. `SYSTEM_TYPE_LAYOUT_KEYS` is the 40 `*-LC-0R` cores used for section goldens.

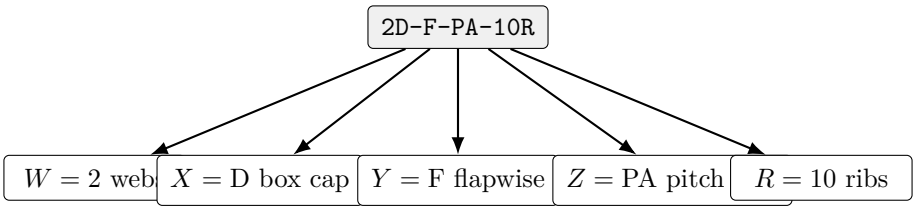


Figure 1: Decode of a catalogue token into station slice (W, X, Y), web freeze Z, and rib count R.

Table 1: Token parts.

Part	Meaning	Examples
W	Shear-web count	`0`, `1`, `2`, ...
X	Spar-cap family (`StructuralFamily`)	`A` none · `B` one band · `C` per-web bands · `D` continuous box
Y	Web orientation	`CN` chord-normal · `F` flapwise (omit when $W = 0$)
Z	Web anchoring	`LC` local chord · `LE` leading edge · `PA` pitch axis
R	Rib count	`0R` none · `10R` ten faces

2 X — spar-cap families

Table 2: Spar-cap families.

X	Description
A	No spar caps: skin, webs, cores (and optional inserts) only.
B	**Single** fixed upper/lower cap pair on **one** chordwise band. Anchor: **pitching** (default $\sim \frac{1}{3}$ chord from LE) or **max_thickness**.
C	**Multiple** discrete upper/lower cap patches — one band per web — merged as `spar_cap_suction_surface` / `spar_cap_pressure_surface`.
D	**Continuous box** spar spanning leftmost to rightmost web (needs $W \geq 2$; no `1D-*`).

3 Y — web orientation

- **CN** (chord-normal): chord-normal in the **untwisted** section.
- **F** (flapwise): vertical in the **twisted** blade frame.

3.1 Implementation note (S / B frames)

CN: the web mid-surface runs chord-normal in the chord (S) frame before section twist. **F**: chord-frame `ShearWebComponent` uses tilted S-frame endpoints so that, after the section's global +twist, the axis is B-frame vertical and meets the same rotated inner-skin clip as skin and caps (`MultiCellSection` flapwise branch — not a `ShearWebComponent` counter-rotation). Spar-cap clipping follows the same chord-vertical vs parallel-strip logic as webs.

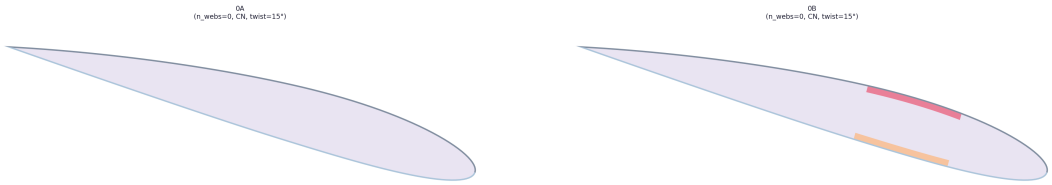
4 Z — web anchoring

The Z letter is `LC`, `LE`, or `PA`. Positions are authored in that mode's native unit (`web_anchoring`). Let $c(z)$ be local chord, $c(0)$ the root chord, s chord fraction from the leading edge, and x blade-frame



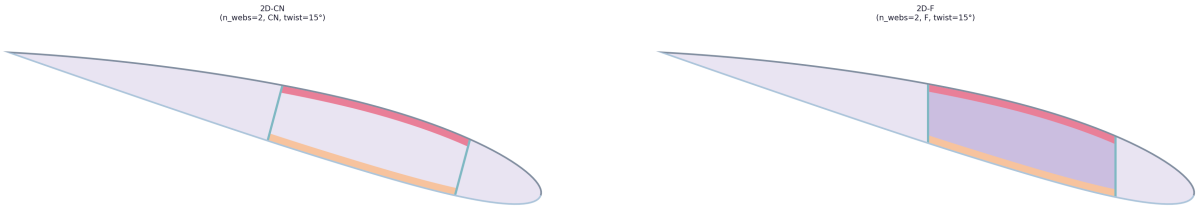
(a) `2A-F`: no caps. **(b)** `2B-F`: one band. **(c)** `2C-F`: per-web bands. **(d)** `2D-F`: continuous box.

Figure 2: Station slices for two flapwise webs, families *A–D* (`examples/demos/figures/structural_family_grids/`).



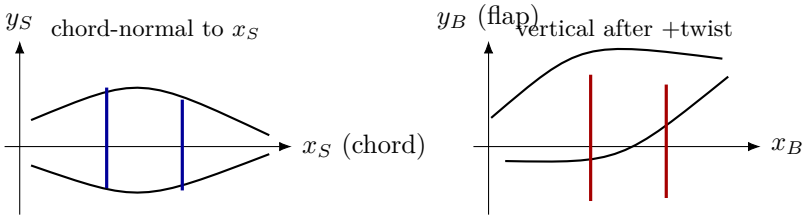
(a) `0A`: skin and cores only. **(b)** `0B`: skin plus one cap band.

Figure 3: Zero-web hosts. Both cores build `MultiCellSection` with empty webs; public tokens still require anchoring, e.g. `0A-LC-0R`.



(a) `2D-CN`: chord-normal webs. **(b)** `2D-F`: flapwise webs.

Figure 4: Same two-web box family under the two orientations.



CN in S (untwisted) **F** in B (twisted)

Figure 5: Orientation Y is not interchangeable between frames: CN is chord-normal in the untwisted S section; F is flapwise-vertical in the twisted B blade frame.

distance from the pitch axis (trailing edge positive).

$$s_k^{\mathrm{LC}}(z) = f_k \in (0,1), \qquad \xi_k^{\mathrm{LE}} = f_k\, c(0), \qquad x_k^{\mathrm{PA}} = \text{const.} \tag{1}$$

Local-chord stations slide with $c(z)$. Leading-edge offsets are metres from the LE and are omitted when ξ_k misses the inner skin. Pitch-axis webs are parallel to the pitch axis on the planform. Up to three primary slots are uniform interiors of the root band; extra slots continue that lattice (first extra at root TE):

$$\Delta x = \frac{x_{\mathrm{TE}}(0) - x_{\mathrm{LE}}(0)}{n_{\mathrm{pri}} + 1}, \quad x_i = x_{\mathrm{LE}}(0) + i\,\Delta x. \tag{2}$$

Primary and secondary bands exist only for PA. Catalogue registry: `catalogue/system_layout.py`. Worked loft: example 09.

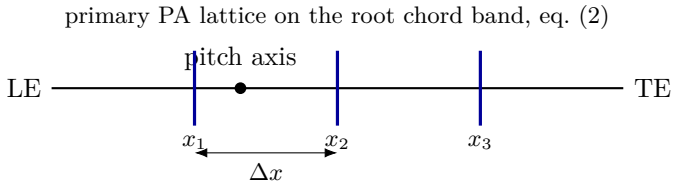
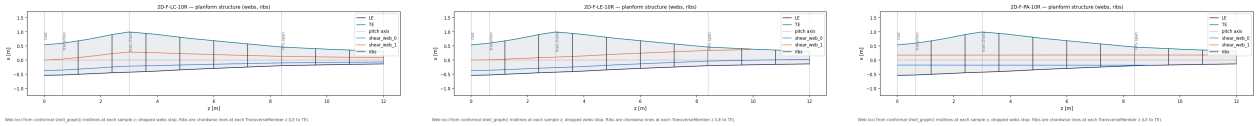


Figure 6: Three primary pitch-axis slots as uniform interiors of the root LE–TE band. Extra slots continue the same lattice (first extra at root TE).



(a) `2D-F-LC-10R`: local chord. **(b)** `2D-F-LE-10R`: leading edge. **(c)** `2D-F-PA-10R`: pitch axis.

Figure 7: Planform structure for the three anchoring letters Z at fixed W=2, X=D, Y=F, R=10 (example 09).

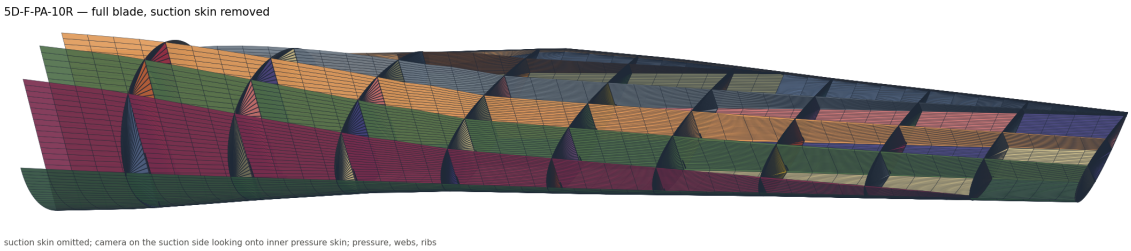
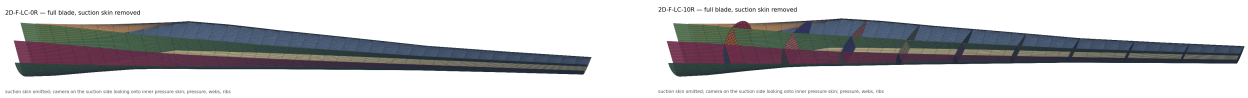


Figure 8: Worked loft `5D-F-PA-10R`: five-web box, flapwise, pitch-axis freeze, ten rib faces.

5 Layout registry (`resolve_system_type`)

Source of truth: `catalogue/system_layout.py` (`SYSTEM_TYPE_KEYS`, `_LAYOUT_REGISTRY`). The cap-family letter of each key is taxonomy X and matches `structural_family` on the spec.

- **0-web cores:** only `0A` and `0B` (no orientation letter). Public tokens still require anchoring: `0A-LC-0R`, `0A-LE-0R`, `0A-LC-10R`. Both cores build `MultiCellSection` with empty webs (family A / B). Catalogue field `geometry_mode="zero_web_catalogue"` is metadata only — it does not select a separate host class.
- **No `0C` / `0D`:** $X = \mathrm{C}$ and $X = \mathrm{D}$ need webs ($W \geq 1$).
- **Multicell cores:** for $W = 1..5$, $X = \mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}$, $Y = \mathrm{CN}, \mathrm{F}$ — all combinations except `1D-CN` and `1D-F`.



(a) `2D-F-LC-0R`: no ribs. **(b)** `2D-F-LC-10R`: ten rib faces.

Figure 9: Rib count R at otherwise identical `2D-F-LC` concepts.

- Total: **40** orientation cores (`SYSTEM_TYPE_LAYOUT_KEYS` is those cores with `-LC-0R`); **120** `SYSTEM_TYPE_KEYS` (each core $\times$ {LC, LE, PA} $\times$ `0R`). Other rib counts (`-10R`) resolve but are not enumerated in the 120.

Table 3: Orientation-core coverage (40 cores). Struck cells are forbidden.

W	A-CN	A-F	B-CN	B-F	C-CN	C-F	D-CN	D-F
0	0A (no Y)		0B (no Y)		—		—	
1	1A-CN	1A-F	1B-CN	1B-F	1C-CN	1C-F	—	—
2	2A-CN	2A-F	2B-CN	2B-F	2C-CN	2C-F	2D-CN	2D-F
3	3A-CN	3A-F	3B-CN	3B-F	3C-CN	3C-F	3D-CN	3D-F
4	4A-CN	4A-F	4B-CN	4B-F	4C-CN	4C-F	4D-CN	4D-F
5	5A-CN	5A-F	5B-CN	5B-F	5C-CN	5C-F	5D-CN	5D-F

6 Spanwise step-down cascade

A token such as `4D-F-PA-10R` is the **blade concept** (station $W/X/Y$, span Z, transverse R). Webs that no longer fit are **dropped**, not slid. The cut keeps that parent token on `SectionCut.layout`. Live web count (W) and cap family (X) are the **station slice** (`station_webs`, `station_family`); they are not a new concept. Display: `4D-F-PA-10R | W=1 X=C dropped=(0,1,2)`.

`effective_layout_for_web_count` / `step_down_cascade` return **assemble recipes** for that slice (how to place caps when N webs survive). Z and R are copied from the parent; they are not rungs.

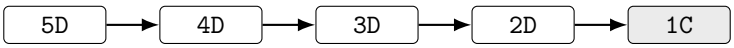
Table 4: Orientation cores by live web count and family (step-down rungs).

W (webs)	X = A	X = B	X = C	X = D
5	5A-CN 5A-F	5B-CN 5B-F	5C-CN 5C-F	5D-CN 5D-F
4	4A-CN 4A-F	4B-CN 4B-F	4C-CN 4C-F	4D-CN 4D-F
3	3A-CN 3A-F	3B-CN 3B-F	3C-CN 3C-F	3D-CN 3D-F
2	2A-CN 2A-F	2B-CN 2B-F	2C-CN 2C-F	2D-CN 2D-F
1	1A-CN 1A-F	1B-CN 1B-F	1C-CN 1C-F	— (no 1D)
0	0A	0B	—	—

W/X floors at **one web**:

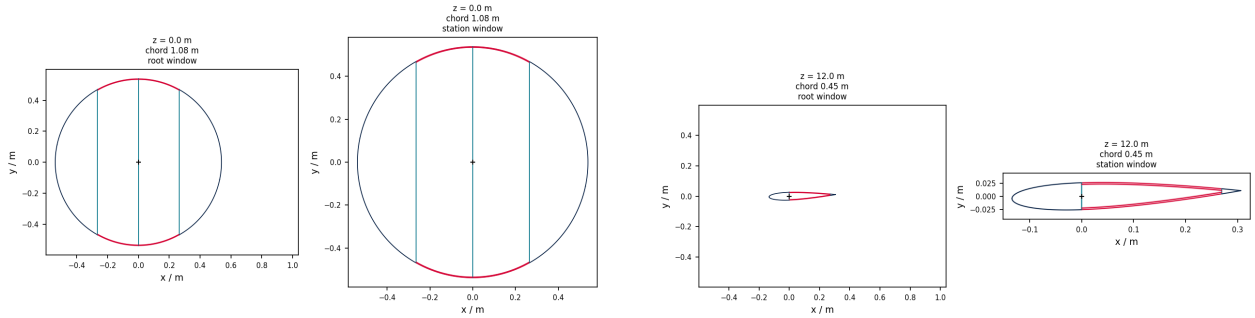
```
A:  nA -> ... -> 1A
B:  nB -> ... -> 1B
C:  nC -> ... -> 1C
D:  nD -> ... -> 2D -> 1C
```

`0A` / `0B` are selectable but not degradation targets. If no web can be anchored, `InfeasibleSectionGeometry` is raised. Family **D** hands to **C** at $W = 1$. Orientation Y is held.



Family D hands to C at $W = 1$. Orientation Y is held.
`0A`/`0B` are selectable but not degradation targets.

Figure 10: Step-down cascade for family D. Families A–C floor at one web of the same letter.



(a) Inboard station: live W still matches the parent.

(b) Tipward station: webs that miss the inner skin are dropped.

Figure 11: Station slice versus concept token. The cut keeps parent `5D-F-PA-10R` on `SectionCut.layout`; realised `station_webs` / `station_family` record the surviving recipe. Display form, e.g. `4D-F-PA-10R | W=1 X=C dropped=(0,1,2)`.

Table 5: Reference axes and loci.

Axis / locus	Role
Pitching axis	Stacking / twist reference; often $\sim \frac{1}{3} \cdot c$ from LE. Not elastic axis or shear centre.
Max-thickness locus	Optional Y = B anchor from sampled `thickness_distribution` on `AirfoilSDF`.
Elastic / centroid / shear centre	Not used for cap placement in v1.
Web-local reference	Y = C caps centred on each web chord station (SG-013).

Table 6: `structural_family` behaviour.

`structural_family`	Behaviour
`"D"` *(default)*	Continuous caps between outer webs
`"A"`	No spar-cap solids
`"B"`	One band; `fixed_cap_anchor`
`"C"`	Union of per-web bands

7 Reference axes and kwargs (advanced)

These axes are not interchangeable.

Kwargs subset:

- `fixed_cap_anchor`
- `pitch_fraction_of_chord_from_le`
- `fixed_cap_chord_half_width`
- `discrete_cap_chord_half_width`

Older docs used $Y = \mathrm{A/B/C}$ with C = box spar; the extended scheme uses B/C for discrete patterns and D for continuous box.

8 Downstream handoff

Gated midlines emit twisted `ShellMidlineStrip` polylines. Prefer `shell_mesh_inputs_from_geometry_result`. Bind orders zones skin → caps → webs; do not assume `panels[i]` aligns with `midlines[i]` when both exist.

9 Caveats (v1)

- Cap placement is not from shear-centre or elastic-axis coordinates.
- Max-thickness anchor uses sampled `thickness_distribution` only.
- `OA` / `OB` are zero-web hosts: skin ± caps midlines hand off without web junctions.

A Demonstration map

Filled slots use existing example and demo figures, or TikZ redraws of schematics that the Markdown already states. Proposed rows are useful later demonstrations; they are not missing assets required to read this PDF.

Table 7: Figure and table slots.

Slot	Where	Status	Asset or note
Token decode	§ Token grammar	filled	Fig. 1 (TikZ)
Token parts	intro	filled	Table 1
Families A–D	§2	filled	Fig. 2, Table 2
Zero-web hosts	§2	filled	Fig. 3
CN vs F stations	§3	filled	Fig. 4
S vs B frames	§3	filled	Fig. 5 (TikZ)
Anchoring eqs. (1)–(2)	§4	filled	display maths
PA lattice	§4	filled	Fig. 6 (TikZ)
LC / LE / PA planforms	§4	filled	Fig. 7
Worked five-web loft	§4	filled	Fig. 8
Rib count $0R$ vs $10R$	§4	filled	Fig. 9
Registry matrix	§5	filled	Table 3
Cascade $W \times X$	§6	filled	Table 4
Family D cascade	§6	filled	Fig. 10 (TikZ)
Station slice vs token	§6	filled	Fig. 11
Axes / kwargs	§7	filled	Tables 5, 6
Spanwise montage of drop events	cascade	proposed	full `cross_sections/01–16` strip with dropped-index labels
Annotated `W=1 X=C` overlay	cascade	proposed	callout on a tip cut matching `dropped=(0,1,2)` exactly
LE miss / omit sequence	$Z = \mathrm{LE}$	proposed	stations where ξ_k leaves the inner skin
Max-thickness B anchor	§7	proposed	overlay of `thickness_distribution` peak on a `B` station
Shell bind order	§8	proposed	skin → caps → webs zone diagram